contato.vnco@gmail.com | Recife, BR | in/vitornegromonte | vitornegromonte | Portfolio

Vitor Negromonte Cabral de Oliveira

Last update: February 19, 2026

Education

Federal University of Pernambuco — Recife
B.Sc Information Systems — 2022–2027

Transferred from B.Sc. Statistics (2022–2025).
Activities: AI League co-founder, Teaching Assistant (Introduction to Deep Learning), Research Assistant.

Work Experience

Lead Backend & Data Architect (FACEPE) — Dec 2025 – Present
Sinapse - Polícia Civil de Pernambuco — **Recife, BR**

Technical lead for the end-to-end criminal intelligence backend, owning architecture, data platform, and governance.

- **System Architecture:** Designed a modular, event-driven monolith (Redis/Celery) to decouple AI workloads and deliver sub-second API performance.
- **Data Platform:** Implemented asynchronous ETL pipelines following Medallion Architecture, leveraging MinIO for immutable storage and PostgreSQL for structured investigative data.
- **Governance:** Established WORM-compliant auditing and automated PII de-identification to ensure LGPD compliance.

Stack: Python (FastAPI), PostgreSQL (pgvector), MinIO, Docker, Redis, Celery.

Machine Learning Engineer — Jan 2025 – Sep 2025
Freelancer — **Remote**

Designed and deployed a full-stack computer vision pipeline for agricultural applications. Managed data collection, dataset creation, image preprocessing, model training, and real-time edge inference.
Stack: Python (PyTorch, OpenCV), Docker, Linux, Shell, Git.

Data Scientist — Nov 2023 – Jul 2024
redduo.ai — **Recife, BR**

Supported business intelligence projects through data analysis and automation tool development. Developed and optimized an NLP pipeline, improving model performance and enabling scalable deployment across production environments.
Stack: Python (Scikit-learn, PyTorch), AWS (Lambda, Cognito), Power BI.

Research Experience

Center for Informatics, Federal University of Pernambuco — Dec 2025 – Present
Undergraduate Researcher — **Recife, BR**

Supervisor: Cleber Zanchettin. Developed a mechanistic interpretability framework for recursive neural architectures. Analyzed dynamical systems properties (Jacobian spectra, attractor basins) and investigated how training objectives shape algorithmic reasoning compared to Transformers.

National Institute of Science and Technology in Software Engineering (INES) — Aug 2023 – May 2024
Undergraduate Research Assistant — **Remote/Recife, BR**

Assisted in the development of quantitative analysis tools and contributed to improving accessibility features in mobile applications aimed at supporting adults on the autism spectrum. Published at SBSI – 2025 and IHC 2024.

Extracurricular

Co-founder & Outreach Director — Mar 2024 – Present
Ligia – Federal University of Pernambuco's Artificial Intelligence Academic League — **Recife, Br**

Lead partnership development, organize AI-focused events, and create educational content to promote AI learning and application across multiple disciplines. The league is a non-profit initiative affiliated with the CIn.AI research group at the Federal University of Pernambuco.

Co-organizer Nov 2025
II Advanced Summer School on Responsible AutoML **Recife, BR**

Event made by the Center for Informatics (UFPE) and the Engineering Faculty of the University of Porto. Coordinated outreach and partnership with interested organizations.

Projects

F1Predict: Formula 1 Race Prediction System – 🔗

Built an end-to-end system for predicting Formula 1 race outcomes, including data collection, feature engineering, model training, and deployment. Developed a REST API with FastAPI and PostgreSQL for data management and model serving. Achieved a 1-second prediction error, demonstrating high real-time accuracy. **Tools: Python, PyTorch, scikit-learn, FastAPI, PostgreSQL.**

MARS: Multi Agent Recommendation System – 🔗

Developed a FastAPI-based system that automates research paper collection from ArXiv using LLM agents to review and filter relevant papers. **Tools: Python (FastAPI).**

Parkinson Diagnosis using Computer Vision: Campus Party'24 Keynote – 🔗

Created a CNN model for Parkinson's disease detection using spiral-drawing images. Outperformed state-of-the-art methods by ≈ 10%, achieving 95% accuracy on the benchmark dataset. **Tools: Python, PyTorch, OpenCV, Optuna, Zeus.**

Computer Vision in Breast Cancer Diagnosis: A Comparative Study with CBIS-DDSM Data – 🔗

Performed a comparative study of multiple CNN architectures for early breast cancer detection using mammography images. **Tools: Python, TensorFlow, OpenCV.**

Publications

[2] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, C. Pereira, A. P. C. Steinmacher, K. Gama, "A Mapping Review to Understand Web and Mobile Apps Accessibility for Adults with Autism", SBSI – 2025

[1] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, G. W. Matias, A. Farias, C. Azul, A. P. Chaves, K. Gama, "A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps", IHC - 2024

Awards & Grants

Cohere Labs Catalyst Grants Program 2025

Teaching

Federal University of Pernambuco (UFPE)

IF867 – Introduction to Deep Learning **2024 – 2026**
- Head Teaching Assistant 2025
- Teaching Assistant 2024

IF866 – Computational Creativity **2024 – 2025**
- Teaching Assistant

Skills & Tools

Programming – Python, R, SQL, LaTeX, Bash.

Data Science & ML – PyTorch (Lightning), JAX, Scikit-Learn, NumPy, Pandas, OpenCV, Weights&Biases, MLFlow.

SWE & DevOps – FastAPI, Django, PostgreSQL, Redis, Docker (Compose), AWS (Lambda, Cognito).

Languages – Portguese (native), English (C1).